AI Model - Installation & Usage

Model

```
Classifier: RandomForest (fallback to rules when unloaded).
Predicts per-task risk level: LOW / MEDIUM / HIGH / CRITICAL.
Endpoint: GET/POST /api/analytics/ai-status/
```

Response payload

```
status, model, features, metrics{accuracy, precision, recall, f1,
confusion_matrix, train_size, test_size}, samples, llm_backend
samples[] = per-task predicted level, confidence, recommended_priority,
            reasons.
```

Live dispatch integration

```
The Genetic Algorithm scheduler combines cost (latency/energy) with
privacy-loss and security-risk so HIGH/CRITICAL tasks avoid weak nodes.
```

Purpose

```
The dataset manager feeds the AI risk-prediction model. Training data:
  datasets/ai_training_10000.csv   (10,000 labelled samples)
Fields: cpu, memory, latency, trust, security_risk, privacy, workload,
        storage_integrity, attack_exposure, risk_level
```

Retrain the model

```
python scripts/train_ai.py        (trains + saves models/risk_model.pkl)
Or in AI Intelligence page press 'Retrain model' (POST /api/analytics/ai-status/).
Validated against a 20% held-out split; accuracy printed after training.
```

Validate / inspect

```
python scripts/validate_dataset.py
python scripts/dataset_check.py
```